Use of adaptive grippers

Python API Usage Instructions

End-of-arm gripper control

1 `set_gripper_state(flag, speed, _type_1=None)`

- **Function** : Let the gripper enter the specified state at a specified speed
- **parameter** :
    - `flag (int)` : 0 - open, 1 - close, 254 - release
    - `speed (int)` : 1 ~ 100
    - `_type_1 (int)` :
        - `1` : Adaptive gripper (adaptive gripper by default)
        - `2` : Five-finger dexterity
        - `3` : Parallel jaws
        - `4` : Flexible gripper
- Return value:
    - 1 - Complete

2 `set_gripper_value(gripper_value, speed, gripper_type=None)`

- **Function** : Allow the gripper to rotate to a specified position at a specified speed
- **parameter** :
    - `gripper_value (int)` : 0 ~ 100
    - `speed (int)` : 1 ~ 100
    - `gripper_type (int)` :
        - `1` : Adaptive gripper (adaptive gripper by default)
        - `2` : Five-finger dexterity
        - `3` : Parallel jaws
        - `4` : Flexible gripper
- Return value:
    - 1 - Complete

3 `set_gripper_calibration()`

- **Function** : Set the current position of the gripper to zero
- Return value:
    - 1 - Complete
-

4 `set_gripper_mode(mode)`

- **Function** : Set the gripper mode
- parameter:
  - `mode` : 0 - transparent transmission. 1 - port mode.
- Return value:
  - 1 - Complete

5 `get_gripper_mode()`

- **Function** : Get the gripper mode
- Return value:
  - `mode` : 0 - transparent transmission. 1 - port mode.

Simple Case

```
from pymycobot.mycobot320 import MyCobot320
import time


# MyCobot320 class initialization requires two parameters:
# The first is the serial port string, such as:
# linux: "/dev/ttyUSB0"
# windows: "COM3"
# The second is the baud rate:
# M5 version: 115200
#
# Example:
# mycobot-M5:
# linux:
# mc = MyCobot320("/dev/ttyUSB0", 115200)
# windows:
# mc = MyCobot320("COM3", 115200)


# Initialize MyCobot320 object
mc = MyCobot320("COM13", 115200)
# Move it to zero position
mc.send_angles([0, 0, 0, 0, 0, 0], 40)
time.sleep(3)
# The professional adaptive gripper needs to be set to transparent transmission
mode before use
mc.set_gripper_mode(0)
time.sleep(1.5)

num = 5
while num > 0:
# Set the gripper state, open the gripper quickly at a speed of 70, parameter 1
represents the professional adaptive gripper type
mc.set_gripper_state(0, 70, 1)
time.sleep(3)
```

```
# Set the gripper state, close the gripper quickly at a speed of 70, parameter 1 represents the professional adaptive gripper type
mc.set_gripper_state(1, 70, 1)
time.sleep(3)
num -= 1

num = 3
while num > 0:
# Set the gripper value to 0, the speed to 70, and parameter 1 represents the professional adaptive gripper type
mc.set_gripper_value(0, 70, 1)
time.sleep(3)
# Set the gripper value to 50, the speed to 70, and parameter 1 represents the professional adaptive gripper type
mc.set_gripper_value(50, 70, 1)
time.sleep(3)
# Set the gripper value to 0, the speed to 70, and parameter 1 represents the professional adaptive gripper type
mc.set_gripper_value(0, 70, 1)
time.sleep(3)
num -= 1

# Get the gripper value
print("gripper value:", mc.get_gripper_value())
```